

Magnetic Pulsations over Antarctica: Results from the Three-Station Triangulation Experiment

S. Gurubaran¹, K. Jeeva¹, M. Sridharan², S. Singh³,
A. Dhar³, A. N. Hanchinal³ and G. Rajaram³

¹Equatorial Geophysical Research Laboratory, Indian Institute of Geomagnetism,
Krishnapuram, Tirunelveli, ²Magnetic Observatory, Indian Institute of
Geomagnetism, Pondicherry, ³Indian Institute of Geomagnetism,
Panvel, Navi Mumbai (e-mail: gurubara@sancharnet.in)

Abstract

Results from the magnetic field observations carried out during the summer campaign of the XX Indian Antarctic Expedition (2000-2001) are presented in this work. Simultaneous data from three locations (Maitri, Dakshin Gangotri and the third near Sigurd Mountain) were made use of in order to identify the ground magnetic signatures of the distant geospace processes that occurred on 24 January 2001. An isolated pulsation event (a westward propagating disturbance) in the Pi3 category was recorded at all the three stations during this period. Triangulation of magnetic records reveals velocities of this waveform to be ~ 1 km/s. The present work clearly highlights the importance of continuing magnetic field measurements in the icy continent, as space weather is an important area that is gaining attention in recent times.

Keywords: Ionosphere, antarctic geomagnetism, solar wind-magnetosphere interactions, currents, magnetic fields.

Introduction

The geographical location of Maitri is unique in that under quiet geomagnetic conditions the station is under the influence of the mid-latitude ionospheric Sq current system. As the magnetic disturbances grow due to the enhanced solar wind-magnetosphere interaction, the atmosphere over Maitri comes increasingly under the auroral oval. The auroral oval expands under such conditions to envelop the sky over Maitri.

Past experiments conducted at the Indian Antarctic station clearly distinguished the disturbed time behaviour of the geomagnetic field variations from their quiet-time characteristics (Kalra et al. 1995, for example). Many important processes occur on the nightside of the Earth wherein the geomagnetic field is stretched to form a long 'tail' by the

streaming charged particle flow from the Sun. The field lines forming the tail are 'energized' whenever the interplanetary magnetic field (IMF) turns southward as this favours magnetic field reconnection on the dayside of the Earth. The reconnected field is swept across towards the nightside. The magnetic field ultimately releases the energy in the form of sporadic bursts in the tail region that manifest as substorms in the ground geomagnetic data (see McPherron 1991, for a review on magnetospheric substorms). The channel in which this energy is routed through is the substorm current wedge that takes a westward path in the high latitude ionosphere. Field-aligned currents constitute the edges of this current pattern (Kisabeth & Rostoker 1973, for example). Simultaneously, the energy extracted from the solar wind may be dissipated in the ionosphere through the directly driven process leading to an enhancement of the convection driven auroral electrojets in the dawn and dusk sectors (Rostoker 1999, for a recent review). This process coincides with the growth phase of a magnetospheric substorm.

The Indian Institute of Geomagnetism (IIG) continues to operate fluxgate magnetometers at the Indian Antarctic station, Maitri (70°45' S, 11°45'E, geographic; 66°S, 53°21'E, geomagnetic). A proton precession magnetometer and a riometer have also been operational for quite some time at this station.

The present work focuses on the electrodynamic conditions prevalent on 24 January 2001 when there was an opportunity to conduct a 3-station triangulation experiment for monitoring the movement of auroral current loops. Intense geomagnetic activity was recorded on this day (ZKp = 26") with moderate storms observed at mid and low latitudes.

Geomagnetic pulsations were recorded during this campaign at three locations separated from each other by 100-200 km distance, which is the scale size of the Hall current loop drifting with the auroral surge. Apart from Maitri, fluxgate magnetometers were operated at Dakshin Gangotri (70°05'S, 12°E) and a third station near Sigurd Mountain (71°20'S, 7°38'E). A triangulation of magnetic field variations at these stations provided information about the velocity of the mobile current systems. Appropriate cross correlation analysis has been performed to quantify the results that compare well with earlier results (Kalra *et al.*, 1998, Rajaram *et al.*, 2002). Using traditional power spectral analysis, the period of the dominant pulsation is identified.

Experiment

During the summer month (January) of 20th Indian Antarctic Expedition, IIG operated three fluxgate magnetometers at the vertices of

a triangular area located in the Schirmacher region of Antarctica. Two of the vertices were at the Indian Antarctic bases, Maitri and Dakshin Gangotri, and the third vertex was located near the Sigurd Mountain of Antarctica. The sides of the triangular setup are 76 km, 160 km and 209 km.

The three fluxgate magnetometers were oriented so as to respond to variations in the X (north-south), Y (east-west) and Z (vertical) components of the Earth's magnetic field variation. The output of the magnetometers was recorded in two modes. The daily variation (DV) was recorded on a chart speed of 3 cm/hr and the sensitivity of the instrument was varied from 10 V Full Scale Deflection (FSD) to 40 V FSD. Micropulsations (MP) in the DV in the period range 30 s to 3000 s (i.e. frequencies from 33 mHz to 0.33 mHz) were separated out through electronic filters, and were recorded on analogue recorders running at a speed of 12 cm/hr. The DV and MP were also recorded in digital form on a solid state data logger, and the data were periodically transferred to floppy diskettes for further computer processing.

The instrument at Dakshin Gangotri (DG) was operated from 22 January till 27 January. Blizzard conditions at this station forced us to terminate the experiment on 27 January. The third station at Sigurd Mountain (SM) camp could be run between 23 January and 27 January. This station too came under severe blizzard conditions halfway through the campaign. When the data were compiled at a later stage, it was found that useful simultaneous magnetic records from all three stations are available only for 24 January. The triangulation method to calculate the velocity of auroral current systems is applied to the data sets obtained on this day (24 January, 2001).

Observations

In Fig. 1 we show the X, Y and Z variations observed at Maitri on 24 January 2001. During the dawn hours, we observe an intense westward electrojet as noted in the X-component. The IMF B_z as seen in the WIND satellite data (not shown here) turned southward around 0200 UT and started turning northward at 0300 UT. A substorm expansive phase is expected to occur under such conditions. This feature was clearly responsible for the ground magnetic signature observed at Maitri during the dawn hours.

The IMF B_z turned southward at 0900 UT on 24 January and continued to remain in this direction since then throughout the observation duration though the solar wind density and pressure were at low levels.

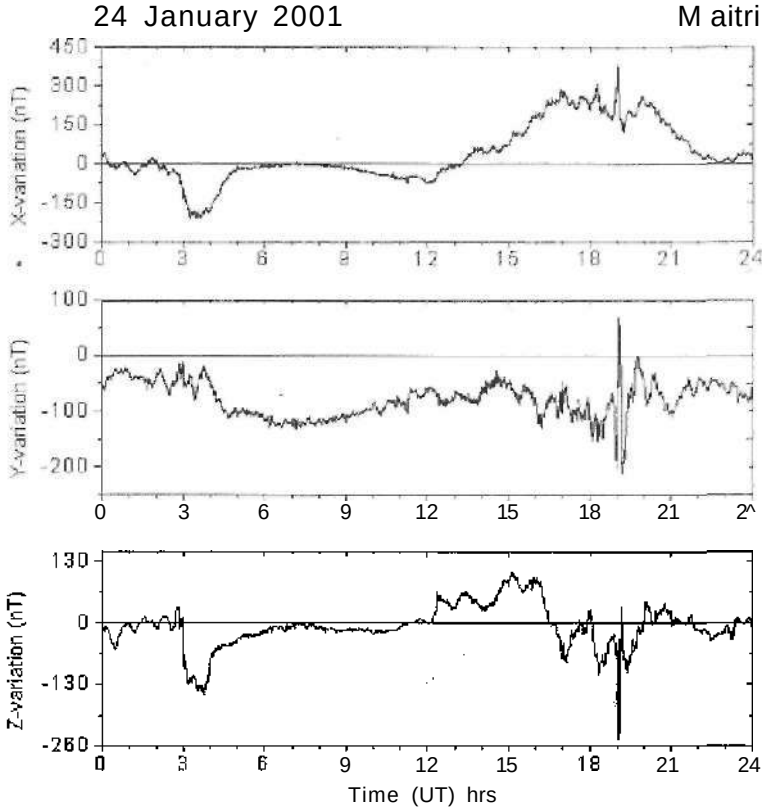


Fig. 1: Geomagnetic field variations in X, Y and Z elements on 24 January 2001 at Maitri.

The v_x component of the solar wind velocity was in the range 400-450 km/s during this interval. An interesting observation noticed in the Maitri records is the presence of quasi-periodic fluctuations in the X-component of the geomagnetic field that commenced at 1200 UT and continued up to 2000 UT.

The physics behind the magnetometer triangulation experiment has been well described in the latest work by Rajaram et al. 2002. In short in a model that assumes disruption of tail current and formation of a localized current wedge the field-aligned currents (FAC) get mapped to the auroral ionosphere forming the boundaries of this wedge. The currents close in the auroral ionosphere. On the western edge strong outward directed FAC corresponding to hot magnetospheric electrons precipitating into the ionosphere, and on the eastern edge weaker inward directed FAC corresponding to cold ionospheric electrons flowing upwards, are

established. In effect the FAC maps into the auroral ionosphere as a westward current wedge. Corresponding to the inward and outward directed FACs are small-scale loops of Hall current which are formed in the auroral ionosphere. This arises due to the generation of radial Pedersen electric fields around the FAC (radially inward for an outward FAC and radially outward for an inward FAC). These small-scale current loops drift eastwards in the dawn sector and westwards in the dusk sector, reflecting the $E \times B$ drift of the ionospheric plasma which originates in the large-scale magnetospheric convection. It is the signature of these drifting ionospheric current loops that is recorded by ground-based magnetometer triangulation experiments at auroral latitudes as time lags in geomagnetic pulsations.

Changes in visible aurora across the midnight sector, namely, the omega bands in the dawnside and the Westward Travelling Surge (WTS), are associated with magnetic pulsations in the Pi3 category (long-period pulsations Ps6 and WTS with time-periods exceeding 150 seconds). The latter features are believed to be associated with the localized small-scale current systems described above (Opgenoorth et al, 1983). Such drifting currents form preferentially during periods of isolated substorms or a series of substorms comprising a storm (Buchert et al., 1990).

The speed and direction of drift of these mobile current systems can be estimated from time lags in their ground geomagnetic signatures. An array of several magnetometers deployed in the magnetic north-south and east-west directions forms a very suitable method for obtaining the velocity of these drifting auroral current systems. Magnetometer arrays have been very effectively used for this purpose in the northern hemisphere (see references cited in Rajaram et al., 2002).

Results

The three fluxgate magnetometers were operated at the vertices of a triangular area located in the vicinity of Maitri (MAI) during the summer campaign period of 2000-2001. The three vertices lie at MAI (70°45'S, 11°45'E), Dakshin Gangotri (70°5'S, 12°E) (DG) and Sigurd Mountain (SM) (71°20'S, 7°38'E). We show in Fig. 2 the geometry of the setup.

We now proceed to make trigonometric calculations that help to determine the speed and direction of the current loops. It may be noted that at the equator, 1 deg. longitude is 111.3 km whereas at 70 deg. latitude, 1 deg. longitude is 38.1 km in distances. The linear distances thus have to be modified considering the spherical Earth. The distances, S_p , S_2 and S_3 , are computed to be 76.6, 159.94 and 209.24 km, respectively.

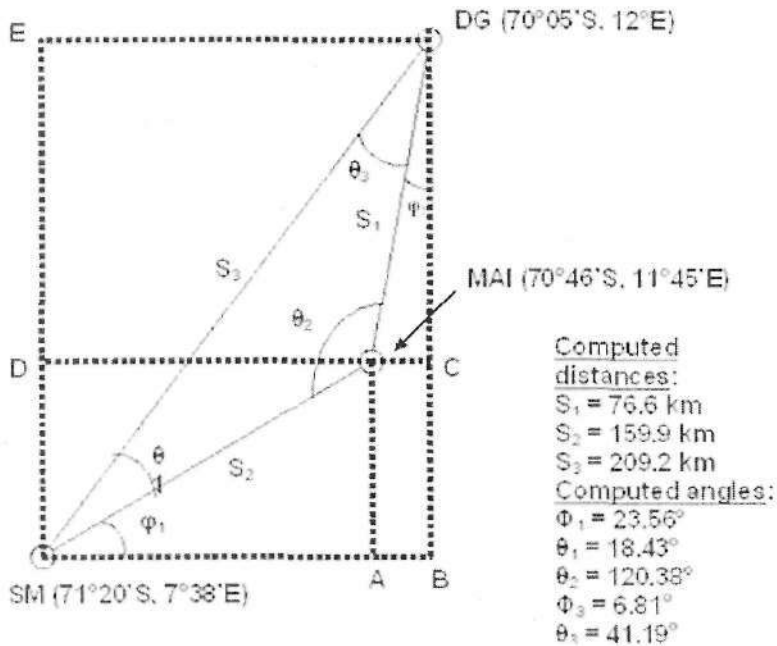


Fig. 2: Geometry of the 3-station triangulation experiment (NOT TO SCALE) :
 Computation of sides and angles forming the triangular shape

We now proceed to calculate the three angles as below.

$$(i) \quad \cos \Phi_1 = (SM-A) / (SM-MAI) = 146.61 / 159.94$$

$$\Phi_1 = 23.56^\circ$$

$$\cos (\theta_1 + \Phi_1) = (SM-B) / (SM-DG)$$

$$SM-DG = 209.24 \text{ km}$$

$$\text{Therefore, } \theta_1 = 18.43^\circ$$

$$(ii) \quad \cos \Phi_3 = (C-DG) / (MAI-DG)$$

Because the distances in the numerator and the denominator are already calculated, we now arrive at $\Phi_3 = 6.81^\circ$

$$\cos (\theta_3 + \Phi_3) = (B-DG) / (SM-DG)$$

Other parameters known, we now arrive at $\theta_3 = 41.19^\circ$

$$(iii) \quad \text{Now } \theta_1 + \theta_2 + \theta_3 = 180^\circ$$

$$\text{Therefore, } \theta_2 = 120.38^\circ$$

Fig. 3 shows the magnetic field variations recorded at the three stations during the time interval 1500 – 0000 UT on 24 January 2001 in the X, Y and Z elements. Though the overall pattern appears to be the

24 January 2001

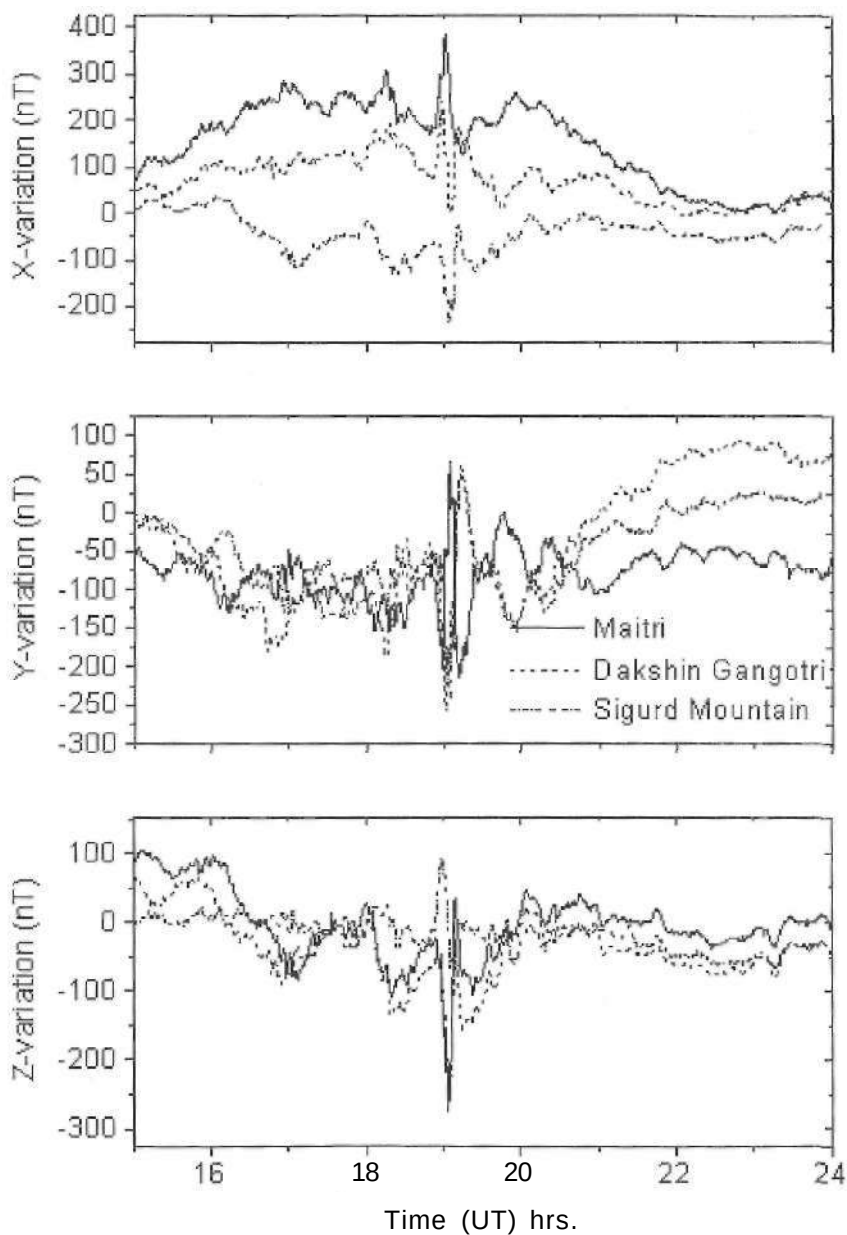


Fig. 3: X, Y and Z field variations recorded at the three stations — Maitri, Dakshin Gangotri and Sigurd Mountain camp, between 1500 and 0000 UT on 24 January 2001.

same over the three stations, there are certain fine features that appear to be different or occurred with a time difference. The X-variation between 1600 and 1800 UT shows a decreasing trend at Sigurd Mountain camp whereas over MAI and DG it shows increasing trends. In the Y-element the positive and the negative pulses observed around 1900 UT clearly appear to be displaced in time between the three stations. Instead of recording a positive bay as noticed at the other two stations, the Y-variation at Maitri reveals a negative bay at around 2000 UT. The differences in Z-variation are relatively smaller except for the sharp negative bursts seen at MAI and DG around 1900 UT that appear as positive variation at Sigurd Mountain.

We shall now examine the pulsation records available in both the digital and analog forms. Fig. 4 shows the X-pulsation amplitude (top)

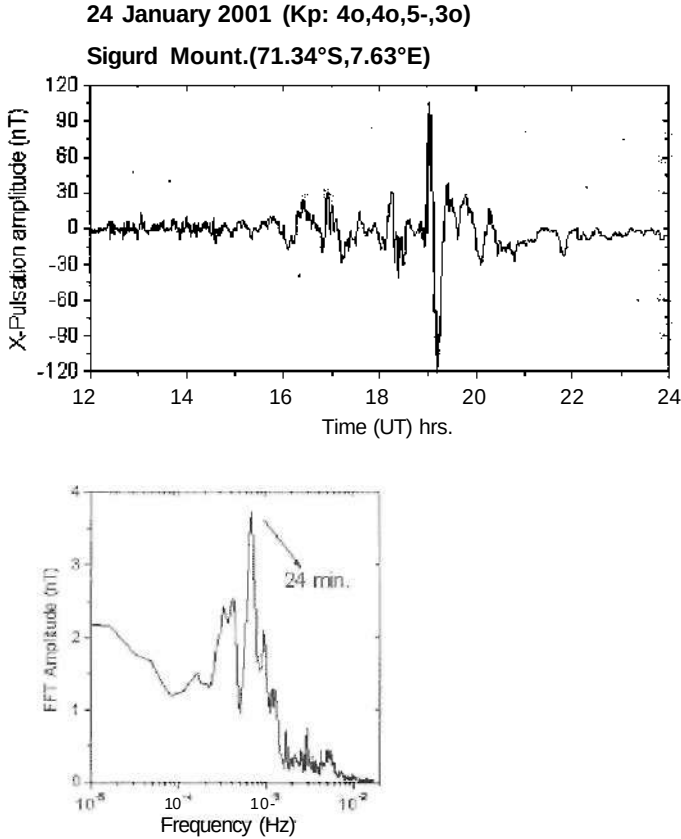


Fig. 4: X-pulsation record (top panel) obtained at Sigurd Mountain camp between 1200 and 0000 UT on 24 January 2001. The bottom panel shows the amplitude spectrum of the pulsations shown in the top panel.

from Sigurd Mountain camp recorded between 1200 and 0000 UT on 24 January 2001. As noted in this figure, a clear isolated event with an amplitude of -120 nT occurring at around 1900 UT has been the dominant feature of this night. The pulsations prior and later to this event were relatively weaker with amplitudes within ± 30 nT. Comparing this with Fig. 1, it may be noted that this large-amplitude event occurring in the dusk sector of the evening auroral ionosphere coincided with the occurrence of the intense eastward electrojet. In other words the magnetic pulsation propagated through the eastward electrojet across the dusk sector of the auroral ionosphere. The 3-station data greatly help to determine the speed and direction of this Pi3 pulsation event. The amplitude spectrum of the X-pulsation records is shown in the bottom panel of Fig. 4. The pulsation event described above is characterized by the 24-minute peak noticed in the amplitude spectrum.

Fig. 5 shows the 1024-sample data set obtained during the time duration 1900 - 1918 UT from the pulsation channels of the X, Y and Z

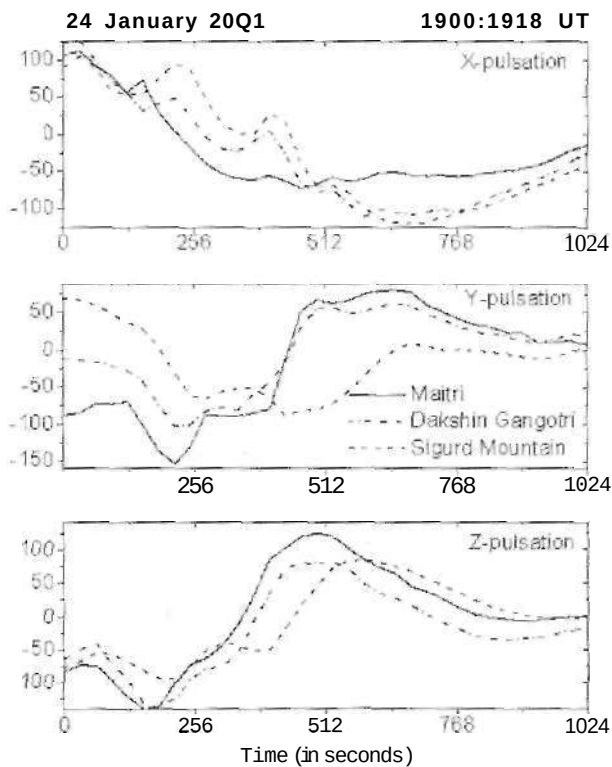


Fig. 5: Pulsation event recorded in the three channels during the period 1900-1918 UT on 24 January 2001. Note the time delays between the three stations.

fluxgate sensors at the three stations. Because the X-pulsation at Maitri is not well-defined, we focus on the Y and Z pulsations both clearly leaving their signatures at all the three stations during this event. There was clearly a delay in the Y and Z pulsations at SG compared to the other two stations. Considering that this station (SG) is geographically west of Maitri we may infer that the pulsation event was propagating toward the west. This is confirmed by the rigorous analysis performed in this work.

From the cross-correlation functions (CCF) for each of the station pairs, it is found that for the event in the Z-element the station MAI leads DG by 12 sec and SM by 79 sec whereas DG leads SM by 67 sec (see Fig. 6). In the Y-element, MAI leads DG by 13 sec and SM by 215 sec.

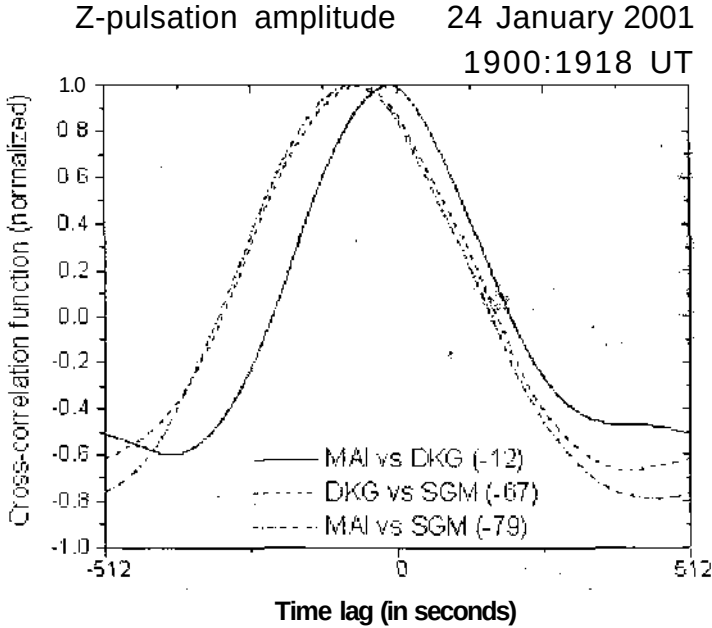


Fig. 6: Cross-correlation functions computed for the three station pairs: Z-pulsations considered for the duration 1900-1918 UT on 24 January 2001.

The triangular form is now rearranged so that it is projected as a right angle triangle as illustrated in Fig. 7.

Z-pulsation event

From the geometry described in Fig. 7, we have

$$\xi = S_2 \cos \Phi_1.$$

$$\Phi_1 = 23.56^\circ \text{ (evaluated earlier).}$$

Therefore, $\xi = 146.61 \text{ km.}$

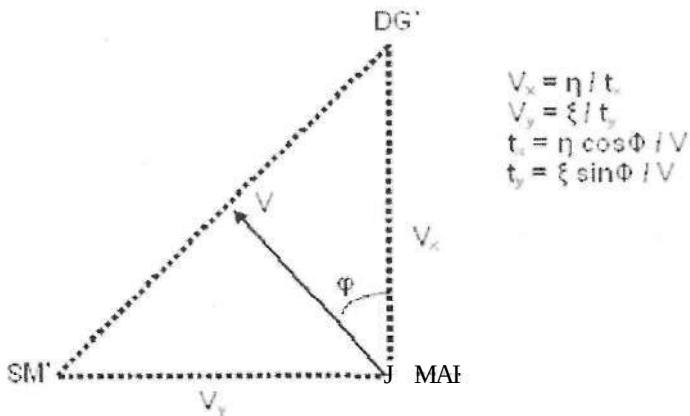
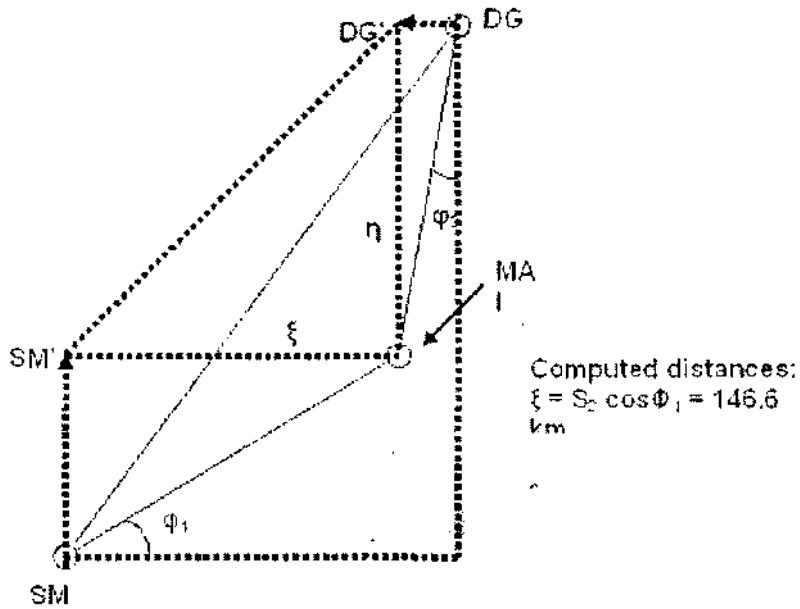


Fig. 7: Geometry of 3-station experiment (NOT TO SCALE): Computation of velocity of the auroral current loop from the time delay measurements

Also, $\eta = S_1 \cos \Phi_3$.

Here $\Phi_3 = 6.81^\circ$.

Hence, $\eta = 76.06 \text{ km.}$

In the x-direction, $v_x = \eta$ (distance) / t_x (time) = 76.06 / 12
= 6.34 km/s.

$$v_y = \xi / t_y = 146.61 / 79 = 1.86 \text{ km/s.}$$

The time displacements in the x and y directions are given by $t_x = \eta \cos \Phi / v$ and $t_y = \xi \sin \Phi / v$.

$$t_x / t_y = \eta / (\xi \tan \Phi).$$

From this expression, we get $\Phi = 73.68^\circ$ and $v = 1.78$ km/s.

This implies that the event propagated towards the north-west direction (73.68° from north in the anti-clockwise direction) with a velocity of 1.78 km/s.

Y-pulsation event

The velocities are computed to be $v_x = \eta / t_x = 5.85$ km/s and

$v_y = \xi / t_y = 0.68$ km/s and the angle is calculated from

$$\tan \theta = \eta t_y / \xi t_x.$$

From these expressions we get, $\Phi = 83.4^\circ$ and $v = 0.67$ km/s.

This implies that the event propagated towards the north-west direction (83.4° from north in the anti-clockwise direction) with a velocity of 0.67 km/s.

Results from the past triangulation experiments conducted by the Indian Institute of Geomagnetism have been reported earlier (Kalra et al., 1998; Rajaram et al., 2002). Experiments carried out during January 1992 and January 1995 had yielded useful results. During January 1992, the third station had been at the Payer's Camp ($71^\circ 51'S$, $14^\circ 29'E$). A magnetic storm occurred on 8 January 1992 with SSC at 0245 UT. There was considerable pulsation activity of the Pi3 type prior to the storm. The velocity of the causative small-scale auroral ionospheric current system for the selected event turned out to be 0.59 km/s with an angle of 102.7° from north direction.

During January 1995, the third station was operated at Orvin camp ($71^\circ 40'S$, $9^\circ 35'E$). Events were selected for the period between 2100 UT on 30 January and 0700 UT on 31 January. The average velocities computed for this period were 1.02 km/s in the eastward direction and 1.58 km/s in the westward direction. The authors of this work (Rajaram et al., 2002) compared their results with those obtained from other stations.

In the present work we identified one pulsation event during the late evening hours of 24 January 2001. It was intense (amplitude exceeding 120 nT) and occurred when the dusk sector over Maitri was under the influence of the convection driven eastward auroral electrojet. Triangulation of magnetic field variations over the three stations revealed a westward propagating disturbance with an average velocity of 1.2 km/s.

Discussion

It has been known for some time that the brightening of an auroral arc is the first indication of the substorm onset. From coordinated optical, radar and magnetometer experiments, Opgenoorth et al. 1983 showed that the arc brightening propagates at speeds exceeding 10 km/s toward the west. Earlier Inhester et al. 1981 analyzed the magnetic perturbations on the ground during the passage of a westward traveling surge (WTS) in the early evening sector in the vicinity of Harang discontinuity. They explained their observations with the combined effects of an upward field-aligned current out of the surge and its associated ionospheric currents. The westward electrojet in the wake of the surge was found to be due to the formation of a Cowling channel that was created by polarization electric fields across a highly conducting channel left behind by the surge. Opgenoorth et al. 1983 attempted to model the three-dimensional current flow in the vicinity of the WTS. The main features of their model were the presence of a localized upward field-aligned current directed out of the head of the surge and fed by a westward electrojet, which is composed of both Hall and Pedersen currents. Secondary ionospheric currents, for example, an anti-clockwise loop of mainly Hall currents around the leading edge of the surge, were found to be responsible for most of the transient effects observed by ground-based magnetometers as the WTS passed overhead.

Referring to the present work, it is possible that the isolated pulsation event (a westward propagating disturbance) recorded around 1900 UT on 24 January 2001 was associated with a surge development originating in the midnight sector east of Maitri. As pointed out by Opgenoorth et al. 1983, in the evening sector the surge associated magnetic and electric disturbance fields constituting the localized small-scale current loops are superimposed on the background fields typical of the corresponding local time sector.

Conclusion

The three-station triangulation experiment performed during the summer campaign of the XX Indian Antarctic Expedition helped to analyze the pulsation events associated with the magnetospheric substorm. The dominant waveform observed at -1900 UT on 24 January had a period of 24 minutes in the Pi3 category of magnetic pulsations. This event has been demonstrated to have originated in the east of Maitri and propagated towards the west with a velocity of - 1 km/s. This was probably associated with a westward traveling surge that is known to be accompanied by field-

aligned current sheets during the expansion phase of a magnetospheric substorm. The localized Hall current loops at the western edge of the surge are believed to generate magnetic pulsation activity that then propagates with a preferred velocity towards the west that can be detected by ground magnetometers.

More work is required to be carried out in order to confirm whether the pulsation activity of the kind observed herein is associated with a westward traveling surge. It would be useful to supplement the ongoing analysis of magnetic pulsations with data from other stations operated in Antarctica.

Finally, with space weather being an important area to be pursued by many international collaborative programmes, the magnetic data sets from Maitri yield a very valuable archive with a wealth of information on major geospace events. In any comprehensive study dealing with various aspects of magnetospheric substorms, experiments of the kind performed during the course of the XX Indian Antarctic Expedition need to be encouraged and should be pursued.

Acknowledgements

The authors gratefully thank the leader of the Expedition, Mr. M. D'Souza, for providing necessary support and encouragement to carry out the three-station experiment. The authors acknowledge the Department of Ocean Development for full travel and logistic support and the Director, Indian Institute of Geomagnetism, for his encouragement on continuing the project on Antarctic geomagnetism.

References

- Butchert, S., Haerendel G. and Baumjohann, W. (1990). A model for the electric fields and currents during a strong Ps6 pulsation event, *J. Geophys. Res.*, 95, 3733-3743.
- Inhester, B., Baumjohann, W., Greenwald, R. A. and Nielsen, E. (1981). Joint two-dimensional observations of ground magnetic and ionospheric electric fields associated with auroral zone currents, 3, Auroral zone currents during the passage of a westward traveling surge, *J. Geophys. Res.*, 155-162.
- Kalra, R., Dhar, A., Unnikrishnan, K., Jeeva, K., Daga, D. M. and Girija Rajaram. (1995). Changes in the auroral electrojet currents inferred from geomagnetic field variation at Maitri and northern conjugate stations, *Eleventh Indian Expedition to Antarctica*, Scientific Report, Department of Ocean Development, Technical Publication No. 9, 87-101.
- Kalra, R., Hanchinal, A. N., Sridharan, M. and Girija Rajaram. (1998). Three station fluxgate magnetometer experiment to determine auroral current velocities over Indian Antarctic

stations, *Fourteenth Indian Expedition to Antarctica*, Scientific Report, Department of Ocean Development, Technical Publication No. 12, pp. 47-55.

Kisabeth, J. L. and Rostoker, G. (1973). Current flow in the auroral loops and surges inferred from ground-based magnetic observations, *J. Geophys. Res.*, 78, 5573.

McPherron, R. L. (1991). Physical processes producing magnetospheric substorms and magnetic storms in *Geomagnetism*, Vol. 4, Edited by J. A. Jacobs, Academic Press.

Oppenoorth, H. J., Pellinen, R. J., Baumjohann, W., Nielson, E., Marklund, G. and Eliasson, L. (1983). Three-dimensional current flow and particle precipitation in a westward traveling surge (observed during the Barium-GEOS rocket experiment), *J. Geophys. Res.*, 88, 3138-3152.

Rajaram, Girija, Hanchinal, A. N., Kalra, R., Unnikrishnan, K., Jeeva, K., Sridharan, M. and Dhar, A. (2002). Velocity of small-scale auroral ionospheric current systems over Indian Antarctic station Maitri, *Proc. Indian Acad. (Earth Planet. ScL)*, 111, 51-62.

Rostoker, G. (1999). The evolving concept of a magnetospheric substorm, *J. Atmos. Solar Terr. Phys.*, 61, 85-100.